



OWNER'S MANUAL & TECHNICAL GUIDE

DELTA ECO



High-Speed Open-Architecture Delta Robot — Installation,
Operation, Safety & Maintenance.

No PLC • No Motion Controller • Just a PC and a Cable

MODEL	DOCUMENT	REVISION	ISSUED
DELTA-ECO	JC2M-DE-OM-001	1.0	May 2026

FRONT MATTER

Document Information

This manual is the official technical reference for the DELTA-ECO delta robot. Keep it accessible to all personnel who install, operate, or service the machine.

Publication Details

Product	DELTA-ECO – Open-Architecture Delta Robot
Document title	Owner's Manual & Technical Guide
Document number	JC2M-DE-OM-001
Revision / status	Rev 1.0 – Released
Issue date	21 May 2026
Prepared by	Team JC2M, AUST
Applies to	DELTA-ECO prototype / competition unit, all configurations

Revision History

REV	DATE	DESCRIPTION OF CHANGE	BY
0.1	Apr 2026	Initial internal draft — structure and specifications.	JC2M
0.9	May 2026	Safety, operation and maintenance sections added; technical review.	JC2M
1.0	May 2026	First released edition.	JC2M

Legal Notice & Disclaimer

© 2026 Team JC2M, American University of Science & Technology. All rights reserved. No part of this document may be reproduced or distributed without written permission of the project team.

DELTA-ECO is a developmental engineering prototype produced for academic and competition purposes. It is not a certified commercial product. Specifications are representative of the as-built unit and are subject to change without notice as the design evolves. The project team accepts no liability for damage or injury resulting from use, modification, or operation outside the conditions described in this manual.

i INFORMATION

Throughout this manual, references to the operator software, control behaviour, and live telemetry correspond to the DELTA-ECO control stack as demonstrated at **[deltaeco.online](#)**.

NAVIGATION

Table of Contents

1	Introduction — manual scope, product overview, intended use	04
2	Safety — alert symbols, hazards, emergency stop, lockout	05
3	Technical Specifications — mechanical, performance, control, environmental	07
4	System Architecture — mechanics, control stack, automation pipeline	09
5	Installation & Commissioning — site requirements, mounting, connections	11
6	Operation — operator interface, start-up, modes, shutdown	12
7	Vision System — camera, calibration, detection workflow	14
8	Maintenance — inspection schedule and procedures	15
9	Troubleshooting — fault diagnosis and corrective action	16
10	Technical Drawings — elevations and motion envelope	17
A	Glossary of Terms	18
B	Quick-Reference Data Sheet	19
C	Warranty, Support & Project Team	20

READ FIRST

Before operating DELTA-ECO for the first time, read **Section 2 — Safety** in full. Operation is restricted to personnel trained on this manual.

1

SECTION ONE

Introduction

— 1.1 About This Manual

This Owner's Manual describes the safe installation, operation, and maintenance of the DELTA-ECO delta robot. It is intended for engineers, integrators, operators, and maintenance technicians. Keep it with the machine for the life of the equipment.

— 1.2 Product Overview

DELTA-ECO is Lebanon's first open-architecture delta robotic arm — a high-speed, three-arm parallel robot built for real industrial material handling. It was developed by Team JC2M at the American University of Science & Technology in partnership with Raymond Feghali Co.

Unlike conventional industrial robots, DELTA-ECO replaces the traditional PLC and dedicated hardware motion controller with a single industrial PC running a real-time Linux kernel. Deterministic motion is delivered over an EtherCAT fieldbus, and AI-based machine vision allows the robot to locate and pick objects dynamically rather than from fixed taught points.

±0.8 mm

REPEATABILITY

9 m/s

MAX SPEED

1 kHz

CONTROL CYCLE

3 kg

MAX PAYLOAD

— 1.3 Intended Use

DELTA-ECO is designed for automated handling of lightweight objects within its defined work envelope, in a controlled industrial or laboratory environment, operated by trained personnel. Supported tasks include:

- **Pick-and-place** — high-frequency transfer of parts with a vacuum or mechanical gripper.
- **Palletizing & depalletizing** — structured row × column × level stacking and destacking.
- **Transfer & sequencing** — coordinated multi-station handling routines.
- **Vision-guided picking** — locating objects on a moving conveyor in real time.

— 1.4 Limitations of Use

⚠ WARNING — DO NOT USE FOR

Payloads above 3 kg; operation outside the defined work envelope or environmental limits; explosive or flammable atmospheres; lifting or transport of persons; or unguarded human-collaborative operation. DELTA-ECO is a developmental unit and must be operated only under qualified supervision.

— 1.5 Manual Conventions

Safety messages in this manual follow a four-level system — **DANGER**, **WARNING**, **CAUTION**, and **NOTICE**. These are defined in Section 2.1. Always observe the action described in each safety message.

2 SECTION TWO

Safety

DANGER — HIGH-SPEED MOVING MACHINERY

DELTA-ECO moves its arm and end-effector at speeds up to 9 m/s. Contact with a moving part can cause serious crushing or impact injury. Never enter the work envelope while the robot is powered and not in a confirmed safe state.

— 2.1 Safety Alert Symbols

This manual uses the following standard safety message levels:



DANGER

Imminent hazard which, if not avoided, will result in death or serious injury.



WARNING

Potential hazard which, if not avoided, could result in death or serious injury.



CAUTION

Potential hazard which, if not avoided, may result in minor or moderate injury.



NOTICE

Practice not related to personal injury; risk of equipment damage or data loss.

— 2.2 General Safety Rules

- Only personnel **trained on this manual** may install, operate, or service the robot.
- The **work envelope must be guarded** — fencing, an interlocked enclosure, or a light curtain — before any automatic operation.
- Keep all body parts, tools, and loose material clear of the envelope during operation.
- Know the location of the **Emergency Stop** before starting the machine (see 2.4).
- Never bypass, defeat, or remove guards, interlocks, or safety devices.
- Do not operate a robot that is damaged, making unusual noise, or behaving unexpectedly.
- Maintain a clean, dry, well-lit, and uncluttered area around the machine.

— 2.3 Residual Hazards

Even when operated correctly, the following hazards remain present:

- **Crush / impact** — the bicep and forearm joints sweep a large volume at high speed.
- **Pinch points** — at the elbow joints and at the moving platform / end-effector.
- **Electrical hazard** — the servo drives and control PC operate at mains voltage.
- **Stored energy** — the arm may settle under gravity when drive power is removed.
- **Unexpected restart** — the robot may move when a fault is cleared or a cycle resumes.

— 2.4 Emergency Stop

The Emergency Stop (E-STOP) immediately removes motion power and brings all three drives to a stop. It is also available as a software control in the operator interface.

WARNING

Test the E-STOP before every operating session. After an E-STOP, identify and correct the cause, confirm the envelope is clear, then re-enable the drives and re-home the robot before resuming. Do **not** use the E-STOP as a routine cycle-stop.

— 2.5 Personal Protective Equipment

- Safety glasses when within the cell area.
- Closed-toe protective footwear.
- No loose clothing, neckties, lanyards, jewellery, or gloves near moving parts.
- Long hair tied back and contained.

— 2.6 Lockout / Tagout (LOTO)

Before any mechanical inspection, cleaning, tool change, or repair inside the work envelope:

- 1 Stop the running cycle from the operator interface and place the robot in **Idle**.
- 2 Engage the **Emergency Stop**.
- 3 Switch off and **lock out the main electrical disconnect**; apply a tag identifying the person working.
- 4 Verify zero energy — confirm drives are unpowered and support the arm against gravity settle.
- 5 Only then enter the work envelope. Reverse the sequence to return to service.

— 2.7 Pre-Operation Checklist

#	CHECK BEFORE EVERY SESSION	OK
1	Work envelope clear of people, tools and loose material.	☐
2	Guarding and interlocks in place and undamaged.	☐
3	Emergency Stop tested and functional.	☐
4	No visible mechanical damage, loose fasteners or cable wear.	☐
5	End-effector securely mounted; payload within 3 kg.	☐
6	Operator software reports EtherCAT: Online and Backend: Running .	☐
7	Robot homed / calibrated before automatic motion.	☐

CAUTION

During first power-up, manual jogging, and calibration the robot may move unexpectedly. Keep all personnel outside the envelope and a hand near the Emergency Stop.

3

SECTION THREE

Technical Specifications

3.1 Mechanical

PARAMETER	VALUE
Configuration	3-arm parallel kinematic (delta)
Arm spacing	120° symmetric
Fixed base radius (R _{base})	120 mm
Bicep (upper-arm) length	320 mm
Forearm rod length	720 mm
Moving platform radius (R _{ee})	70 mm
Mounting orientation	Overhead, end-effector down

3.2 Performance

PARAMETER	VALUE
Maximum payload	3 kg
Repeatability	±0.8 mm
Positioning resolution	±0.799 mm
Maximum end-effector speed	9 000 mm/s (9 m/s)
Peak pick rate	120 picks/min
Control cycle time	1 ms (1 kHz)

3.3 Motion Envelope

AXIS / JOINT	RANGE
Cartesian X travel	-350 mm to +350 mm
Cartesian Y travel	-350 mm to +350 mm
Cartesian Z travel	-482 mm to -957 mm
Joint θ1 range	-37.9° to +92.9°
Joint θ2 range	-45.3° to +94.2°
Joint θ3 range	-45.3° to +94.2°

i NOTICE

The Z axis is referenced from the fixed base plate; all Z values are negative (below the base). Soft limits in the motion software prevent commands beyond the ranges above — do not attempt to defeat them.

— 3.4 Control & Software

PARAMETER	VALUE
Controller	Industrial PC, PREEMPT_RT Linux kernel
Hardware motion controller / PLC	None – motion is software-defined
Fieldbus	EtherCAT, distributed-clock synchronised
Fieldbus master	IgH EtherLab EtherCAT Master
Motion software	Custom C motion daemon (IK + trajectory)
Servo drives	3, Cyclic Synchronous Position (CSP) mode
Vision	Logitech C170 camera + YOLO object detection
Operator interface	Custom GUI – 9 operating screens

— 3.5 Real-Time Performance

PARAMETER	VALUE
Kernel latency (jitter), max	$\leq 0.8 \text{ ms}$
EtherCAT distributed-clock drift	$\approx 12 \text{ ns}$
Real-time CPU load (typical)	$\approx 24 \%$
EtherCAT synchronisation state	Locked during operation

— 3.6 Operating Environment

PARAMETER	RECOMMENDED
Installation	Indoor, dry, dust-controlled
Ambient temperature	5 °C to 40 °C
Relative humidity	20 % to 80 %, non-condensing
Mounting structure	Rigid, level, vibration-free frame
Lighting (for vision)	Stable, even, glare-free

— 3.7 Site & Commissioning Data

To be completed by the integrator during installation. Values depend on the installed servo drives and power supply.

PARAMETER	RECORDED VALUE
Mains supply voltage / phase	–
Supply frequency	–

PARAMETER	RECORDED VALUE
Full-load current	—
Control / logic voltage	—
End-effector type & supply	—
Total installed mass	—

4

SECTION FOUR

System Architecture

4.1 Mechanical Structure

DELTA-ECO uses a three-arm parallel (delta) kinematic structure. Three servo motors are fixed to the stationary base plate, 120° apart. Each motor drives a rigid **bicep**, which connects through a parallelogram pair of **forearm rods** to a small **moving platform** that carries the end-effector.

Because all three motors are mounted on the fixed base, the moving mass is very low — this is what allows a delta robot to reach high speed and acceleration while keeping the platform horizontal throughout its travel.

4.2 Control Architecture

The defining feature of DELTA-ECO is its **open control architecture**. A traditional delta cell uses a proprietary PLC plus a dedicated hardware motion controller. DELTA-ECO replaces that entire rack with a single industrial PC.

LAYER	TRADITIONAL CELL	DELTA-ECO
Controller	Proprietary PLC / ASIC	Industrial Linux PC
Programming	Ladder / Structured Text	Native C / CSP commands
Cycle time	10-20 ms	1 ms, jitter-free
Logic access	Vendor locked	Fully open

A real-time Linux kernel (PREEMPT_RT) with isolated CPU cores hosts the IgH EtherLab EtherCAT master. The DELTA-ECO motion daemon runs a deterministic 1 kHz loop: it reads encoder feedback, solves the inverse kinematics, and streams cyclic position commands to all three servo drives over EtherCAT — with sub-millisecond, distributed-clock-synchronised behaviour.

RESULT

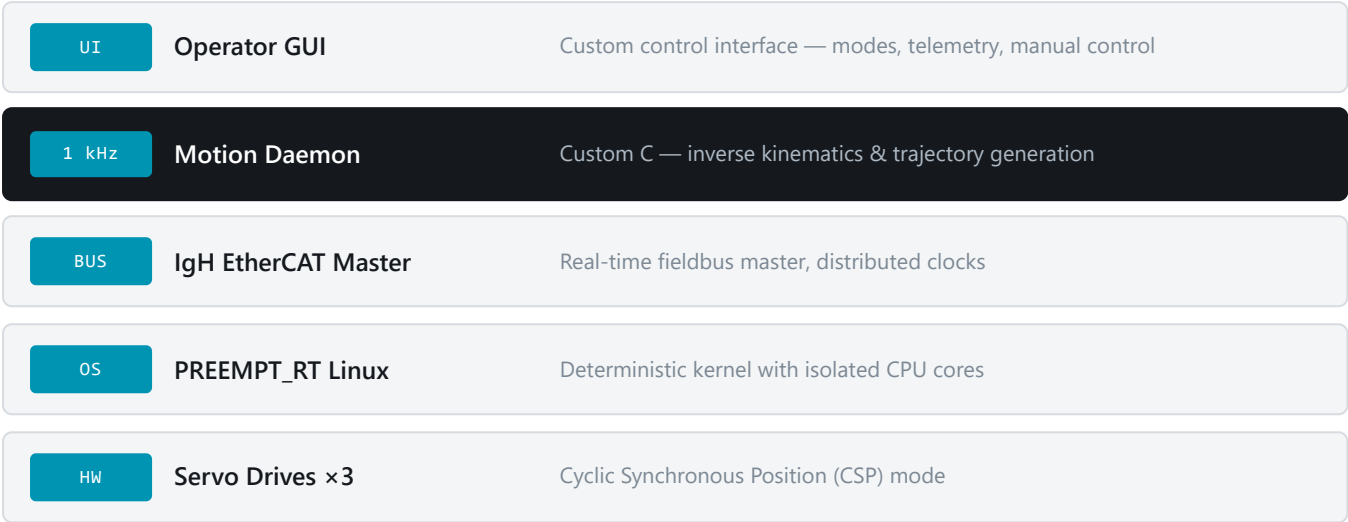
What used to require a rack of dedicated hardware now runs on a single machine — with no vendor lock-in and no licence dependency.

4.3 Inverse Kinematics

Every Cartesian target (X, Y, Z) is converted by the custom C solver into the three joint angles θ_1 , θ_2 , θ_3 required to place the platform at that point. The solver runs inside the 1 kHz loop, so the robot can be commanded directly in workspace coordinates — including coordinates supplied live by the vision system.

— 4.4 Software Stack

The control software is organised as five layers, from the operator-facing interface down to the servo hardware:



— 4.5 Automation Pipeline

From sensor input to executed motion, five deterministic stages run on every control cycle:



— 4.6 Safety & Protective Functions

- **Soft joint limits** — commands beyond the joint ranges in 3.3 are rejected.
- **Torque limits** — drive torque is capped to protect the mechanism and payload.
- **Following-error monitoring** — excessive position error triggers a controlled stop.
- **Jitter monitoring** — loss of real-time determinism is detected and reported.
- **Emergency-stop integration** — hardware and software E-STOP remove motion power.

5

SECTION FIVE

Installation & Commissioning

⚠ WARNING

Installation and electrical connection must be carried out by qualified personnel only. Keep the robot de-energised until commissioning is complete.

— 5.1 Unpacking & Inspection

- Inspect for transit damage before installation. Report any damage before proceeding.
- Confirm all three arms, the moving platform, the end-effector, the control PC, and cabling are present.
- Handle the forearm rods and ball joints with care — they are precision components.

— 5.2 Site Requirements

- A **rigid, level overhead frame** able to carry the robot mass and dynamic reaction loads.
- Clearance for the full work envelope (see Section 10) plus a guarded safety margin.
- Indoor environment within the limits of Section 3.6.
- A clean, stable power supply and a lockable main disconnect.

— 5.3 Installation Procedure

- 1 Mount the **fixed base plate** to the overhead frame, level, with the end-effector facing down. Torque all fasteners to specification.
- 2 Confirm the three biceps and forearm rods move freely through their full range without binding.
- 3 Fit and secure the **end-effector** to the moving platform; route its supply line with proper strain relief.
- 4 Connect the three servo motors to their drives; connect the drives in the **EtherCAT** chain to the control PC.
- 5 Connect the **vision camera** to the control PC and mount it over the work area (see Section 7).
- 6 Wire the **Emergency Stop** and any guarding interlocks. Verify mains earth/ground continuity.
- 7 Install guarding around the full work envelope before any powered motion.

— 5.4 First Power-Up & Commissioning

- 1 With the envelope clear, apply power and boot the control PC.
- 2 Launch the operator software and start the backend. Confirm **EtherCAT: Online** and all drives are detected.
- 3 Run **Calibrate** to home the robot. Verify the reported joint angles are sensible.
- 4 Jog each axis slowly in Manual mode and confirm correct, smooth motion within the envelope.
- 5 Test the Emergency Stop under motion. Record the commissioning data in Section 3.7.

6

SECTION SIX

Operation

— 6.1 The Operator Interface

DELTA-ECO is operated entirely from the custom control software, which presents the robot as a real industrial control station. The interface provides nine screens:

SCREEN	FUNCTION
Dashboard	System status, cycle count, tool position, loop jitter.
Manual Control	Direct Cartesian jogging of the end-effector (X / Y / Z).
Pick and Place	Single pick → place transfer routines.
Palletizing	Fill a row × column × level grid from one pick point.
Depalletize	Vision-guided destacking of mixed units.
Transfer	Station-to-station coordinated handling.
Sequence	Multi-step routines chained into one program.
Vision	Live camera feed and YOLO detection status.
Settings	Calibration, limits, and system configuration.

A persistent status bar shows **Backend**, **EtherCAT**, and **Robot** state, with **Calibrate** and **Stop Backend** controls. A right-side control panel keeps the **Emergency Stop** and live drive states visible at all times. For each of the three drives the software reports state, status word, mode, actual and command position, position error, velocity, and acceleration.

— 6.2 Start-Up Procedure

- 1 Complete the **Pre-Operation Checklist** (Section 2.7). Confirm the envelope is clear.
- 2 Power on the control enclosure and boot the PC; launch the operator software.
- 3 Start the backend — wait for **Backend: Running** and **EtherCAT: Online**.
- 4 Press **Calibrate** to home the robot. The drives transition from *Not Ready* to *Ready*.
- 5 Select the required operating mode. The robot is now ready for production.

⚠ CAUTION

The robot will move during calibration. Ensure the envelope is clear and guarded before pressing **Calibrate**.

— 6.3 Operating Modes

Manual Control

Jog the end-effector directly along X, Y, and Z. The live tool position is displayed continuously. Use Manual mode for setup, teaching reference points, and verifying motion. Always jog at reduced speed near limits.

Pick-and-Place

The robot transfers an object from a pick point to a place point using the gripper. Pick and place coordinates can be entered numerically or captured from the current position.

Palletizing & Depalletizing

Define a **Rows × Columns × Levels** grid, the item dimensions, and X/Y spacing. From a single taught pick point the robot fills (or, in Depalletize, empties) the grid automatically with consistent spacing. Hand-captured manual place points are also supported.

Transfer & Sequence

Transfer performs coordinated station-to-station handling. Sequence chains multiple steps — picks, places, transfers — into a single repeatable program for multi-step routines.

— 6.4 During Operation

- Monitor the Dashboard — cycle count, tool position, and loop jitter should remain stable.
- Keep all personnel outside the guarded envelope while the robot is in automatic motion.
- Stop immediately if you observe unusual noise, vibration, rising position error, or a jitter warning.
- Do not exceed the 3 kg payload or command targets outside the work envelope.

— 6.5 Shutdown Procedure

- 1 Allow the current cycle to finish; place the robot in **Idle**.
- 2 Stop the backend from the operator software.
- 3 Shut down the control PC.
- 4 Switch off the control enclosure at the main disconnect.

i NOTICE

For an immediate stop at any time, use the Emergency Stop. After an E-STOP, follow the recovery steps in Section 2.4 before resuming.

7

SECTION SEVEN

Vision System

DELTA-ECO sees before it moves. The vision system lets the robot adapt to object position dynamically, instead of relying on fixed pick points or mechanical guides.

— 7.1 Components

ITEM	DETAIL
Camera	Logitech C170
Detection model	YOLO object detection
Processing frame size	1280 × 720
Output	Object class + XYZ pick coordinate

— 7.2 How It Works

- 1 The camera captures the live conveyor scene as calibrated image data.
- 2 The YOLO model **localises and classifies** each target part directly from the running scene.
- 3 Each detection is converted into an **XYZ coordinate** in the robot's workspace.
- 4 The coordinate is fed straight into the inverse-kinematics solver for precision targeting.

— 7.3 Camera Mounting & Calibration

- Mount the camera rigidly above the work area with an unobstructed view of the conveyor.
- Provide stable, even lighting — avoid glare, shadows, and changing ambient light.
- Calibrate the camera-to-robot coordinate mapping so detected pixels translate to accurate XYZ.
- Re-verify calibration after any change to camera position, lens focus, or lighting.

— 7.4 Operating Notes

i NOTICE

Detection accuracy depends on lighting and a clean lens. If detection confidence drops or parts are missed, check the camera connection, clean the lens, and confirm lighting before adjusting the model. Verify the vision-derived pick points stay inside the work envelope (Section 3.3).

BENEFIT

Because objects become robot targets directly, the line does not need fixed fixturing — the robot adapts to where the part actually is.

8

SECTION EIGHT

Maintenance

⚠ WARNING

Apply the Lockout / Tagout procedure (Section 2.6) before any maintenance inside the work envelope. Never service the robot while it is powered.

— 8.1 Maintenance Schedule

INTERVAL	TASK
Each shift / daily	Visual inspection; confirm work area clear; test Emergency Stop; listen for unusual noise or vibration; check cable dressing.
Weekly	Inspect ball joints and rod ends for play; check end-effector mounting; clean the camera lens; verify guarding and interlocks.
Monthly	Check bicep and base fastener torque; inspect forearm rods and retention springs; inspect EtherCAT and motor cabling for wear.
Semi-annual	Full fastener torque check; bearing and joint inspection; recalibrate the robot; recalibrate the vision system.
Annual	Review drive logs and jitter history; update / back up software configuration; replace worn ball joints or rods as required.

— 8.2 Inspection Points

- **Ball joints & forearm rods** — the highest-wear parts. Replace any joint with detectable play or any bent rod.
- **Fasteners** — high acceleration loosens hardware over time. Re-torque to specification.
- **Cabling** — check the EtherCAT chain and motor leads for chafing, especially at flex points.
- **End-effector** — confirm secure mounting and a functioning gripper / vacuum supply.
- **Camera** — keep the lens clean; confirm the mount has not shifted.

— 8.3 After Maintenance

Following any mechanical work, re-home the robot with **Calibrate**, jog each axis slowly in Manual mode, verify the Emergency Stop, and run a low-speed test cycle before returning the machine to production.

i NOTICE

Record completed maintenance, replaced parts, and recalibration dates in a machine log. A consistent log is the fastest route to diagnosing future faults.

9

SECTION NINE

Troubleshooting

Use the table below for first-line diagnosis. If a fault persists after the corrective action, stop the machine, apply Lockout / Tagout, and contact the project team.

SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
EtherCAT shows <i>Offline</i>	Cable unplugged, drive unpowered, or master not started.	Check the fieldbus cabling and drive power; restart the backend.
Drive shows <i>Not Ready</i>	Drives not enabled, or a fault is latched.	Clear faults, run Calibrate , confirm drives transition to <i>Ready</i> .
Robot will not move in Manual	Not homed, target outside soft limits, or E-STOP engaged.	Release the E-STOP, calibrate, command a target inside the envelope.
Rising position / following error	Excess payload, speed too high, or a mechanical bind.	Reduce payload or speed; inspect joints and rods for binding.
Vision not detecting parts	Camera disconnected, poor lighting, or model not loaded.	Check the USB connection, improve lighting, clean the lens, reload the model.
High jitter warning	Background CPU load, or core isolation lost.	Close non-real-time processes; verify kernel core isolation.
Unexpected stop mid-cycle	E-STOP, safety interlock, or a protective fault.	Inspect guarding interlocks; review the system log; clear and resume per 2.4.
Unusual noise or vibration	Loose fastener, worn ball joint, or end-effector imbalance.	Stop, apply LOTO, inspect fasteners and joints; replace worn parts.
Inaccurate pick position	Calibration drift, or vision mapping out of date.	Re-home the robot; recalibrate the camera-to-robot mapping.

⚠ WARNING

Never attempt to diagnose a fault by reaching into the work envelope while the robot is powered. De-energise and apply Lockout / Tagout first.

— 9.1 Diagnostic Resources

The operator software exposes a live system log, per-drive status words, real-time jitter and CPU-load telemetry, and the EtherCAT synchronisation state. Capture this information before contacting support — it greatly speeds up diagnosis.

10

SECTION TEN

Technical Drawings

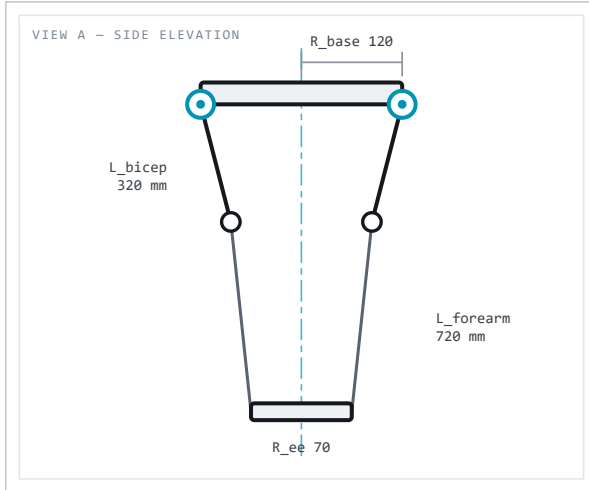


FIG. 1 – SIDE ELEVATION

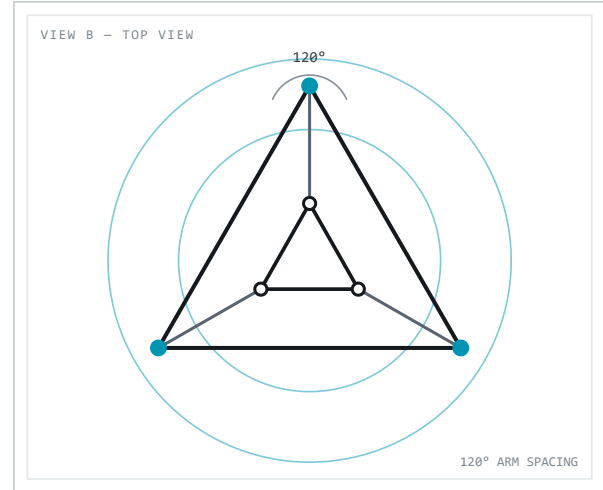


FIG. 2 – TOP VIEW / ARM LAYOUT

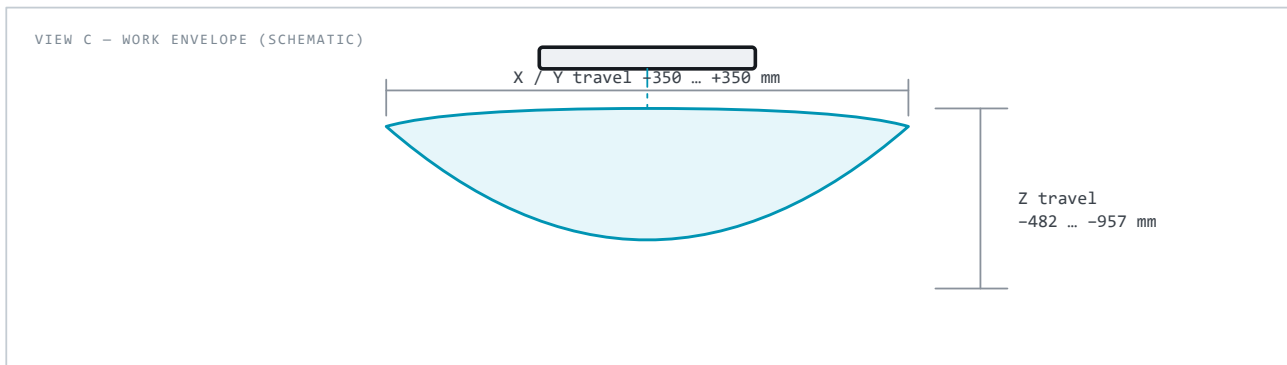


FIG. 3 – REACHABLE WORK ENVELOPE

NOTE

Drawings are schematic and not to scale. Dimensions in millimetres. Refer to Section 3 for the complete specification.

APPENDIX A

Glossary of Terms

Delta robot

A parallel-kinematic robot in which three arms, all driven from a fixed base, support a single moving platform — giving very high speed and low moving mass.

Bicep

The rigid upper-arm link driven directly by a base-mounted servo motor.

Forearm

The pair of parallel rods linking each bicep to the moving platform; keeps the platform level throughout travel.

End-effector

The tool mounted on the moving platform — a vacuum or mechanical gripper — that handles the payload.

Work envelope

The full volume the end-effector can reach. See Section 3.3 and Fig. 3.

Repeatability

How closely the robot returns to the same point on repeated cycles. DELTA-ECO: ± 0.8 mm.

IK — Inverse Kinematics

The computation that converts a Cartesian target (X, Y, Z) into the joint angles needed to reach it.

EtherCAT

A real-time industrial fieldbus used to connect the controller to the servo drives with deterministic timing.

Distributed Clocks

An EtherCAT mechanism that synchronises all devices on the bus to a common time base.

CSP — Cyclic Synchronous Position

A servo-drive control mode in which the controller sends a fresh target position every cycle.

PREEMPT_RT

A real-time patch for the Linux kernel that makes task timing deterministic enough for motion control.

IgH EtherLab

The open-source software that turns a Linux PC into an EtherCAT fieldbus master.

YOLO

A real-time AI object-detection method that locates and classifies objects in a camera image.

Jitter

Cycle-to-cycle variation in the timing of the control loop. Lower jitter means more deterministic motion.

HMI — Human-Machine Interface

The operator software through which the robot is configured, monitored, and controlled.

LOTO — Lockout / Tagout

The procedure for isolating and securing machine energy before maintenance. See Section 2.6.

APPENDIX B

Quick-Reference Data Sheet

A single-page summary of key data. For full detail and context, refer to the numbered sections.

— Performance

Max payload	3 kg
Repeatability	±0.8 mm
Resolution	±0.799 mm
Max speed	9 m/s
Peak pick rate	120 picks/min
Control cycle	1 ms (1 kHz)

— Mechanical

Base radius	120 mm
Bicep length	320 mm
Forearm length	720 mm
Platform radius	70 mm

— Work Envelope

X / Y	±350 mm
Z	−482 to −957 mm
θ1	−37.9° / +92.9°
θ2, θ3	−45.3° / +94.2°

— Control

Controller	Linux PC
Fieldbus	EtherCAT
Drives	3 × CSP
Vision	C170 + YOLO

— Emergency Procedure

⚠ IN AN EMERGENCY

Press the **EMERGENCY STOP**. Do not enter the work envelope until the robot is confirmed stopped and de-energised. Identify and correct the cause, confirm the envelope is clear, then re-enable the drives and re-home before resuming — see Section 2.4.

— Daily Start-Up — At a Glance

1. Complete the Pre-Operation Checklist (2.7) — envelope clear, E-STOP tested.
2. Power on, boot the PC, launch the operator software.
3. Start the backend — wait for **EtherCAT: Online**.
4. Press **Calibrate** to home the robot.
5. Select the operating mode and begin production.

KEEP VISIBLE

Consider posting this page at the operator station for quick day-to-day reference.

APPENDIX C

Warranty, Support & Project Team

— Project & Support

DELTA-ECO was designed, built, and integrated in Lebanon by Team JC2M at the American University of Science & Technology, in partnership with Raymond Feghali Co. For a live demonstration, a technical deep-dive, or a partnership conversation, contact the project team.

Project	DELTA-ECO – Open-Architecture Delta Robot
Team	JC2M, AUST
Email	team@deltaeco.online
Web	deltaeco.online
Based in	Beirut, Lebanon

— Project Team



Mounir Rizkallah
Mechanical Design & Kinematics
Mechatronics Engineer



James Youna
Control Systems & EtherCAT
Computer & Communication Eng.



Majd Hajj Ali Sleiman
Computer Vision & AI Integration
Computer & Communication Eng.



Cirine Radwan
HMI & Software Integration
Computer & Communication Eng.

— Academic Advisors

Mr. Michel Owayjan	Project Advisor
Mr. Mohammad Ballouz	Co-Advisor
Mr. Marcel Khalaf	Technical Advisor
Mr. Fadi Elias Sleiman	Technical Advisor

— Warranty & Liability Statement

DELTA-ECO is a developmental engineering prototype produced for academic and competition purposes; it is not a certified commercial product and carries no commercial warranty. It must be operated only by trained personnel, under qualified supervision, and strictly within the conditions of this manual. The project team accepts no liability for damage or injury arising from use, modification, or operation outside those conditions.

KEEP

Retain this manual with the machine. Replace it if it becomes damaged or illegible.



DELTAECO

OPEN-ARCHITECTURE ROBOTICS • DESIGNED & OWNED IN LEBANON

WEB

deltaeco.online

EMAIL

team@deltaeco.online

BUILT IN

Beirut, Lebanon